

The Concepts of Business Intelligence

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- Introduction
- Consolidating Data from Multiple Sources
- Supporting Different Types of Users
- Identifying Elements to Support Analysis

Business Intelligence (BI)

- The processes, technologies and tools needed to turn data into information and information into knowledge and knowledge into action plans that drive profitable business action.
- BI encompasses data warehousing, business analytics and knowledge management.

OLAP vs. Business Intelligence

Online analytical processing, or OLAP

- It is an approach to quickly answer multi-dimensional analytical queries.
- OLAP is part of the broader category of [business intelligence](#), which also encompasses reporting, data mining, and analytics.

The Challenges of Building BI Solutions

- There are several issues inherent to any BI project:
 - Data exists in multiple places
 - Data is not formatted to support complex analysis
 - Different kinds of workers have different data needs
 - What data should be examined and in what detail
 - How will users interact with that data

Consolidation of Data

- The process of consolidating data means moving it, making it consistent, and cleaning up the data as much as possible
 - Data is frequently stored in different formats
 - Data is frequently inconsistent between sources
 - Data may be dirty
 - Internally inconsistent or missing values

Disparate Data

- Data in a variety of locations and formats:
 - Relational databases (operational data systems)
 - XML files
 - Desktop databases
 - Microsoft[®] Excel[™] spreadsheets
- The data may also be in databases on different operating system and hardware platforms

Inconsistent Data

- Data may be inconsistent
 - Two plants might have different part numbers for the same physical part
 - To represent True and False, one system may use 1 and 0, while another system may use T and F
 - Data stored in different countries will likely store sales in their local currency
 - These sales must be converted to a common currency

Data Quality Issues

- Clean data facilitates more accurate analysis
- Many data entry systems allow free-form data entry of text values
 - For example, the same city might be entered as Blore, Bangalore, and Bangaluru
- Routines to clean up data need to take into account all possible variations of bad data

Extraction, Transformation, and Loading (ETL)

- The process of data consolidation is often called Extraction, Transformation, and Loading (ETL)
 - The ETL process *extracts* data from the various source systems
 - Data is then *transformed* to make it consistent and improve data quality
 - The consolidated, consistent, and cleaned data is then *loaded* into a data repository
- Developing the ETL process often consumes 80% of the development time

Extraction, Transformation, and Loading (ETL) Tools

- Some ETL Tools
 - Apache Airflow
 - Oracle Data Integrator (ODI)
 - Informatica
 - IBM Ascential
 - Abinitio

Business Issues with Data Consolidation

- Business users must drive what should be in the data warehouse
- Someone in the business must decide how to consolidate inconsistent data
 - If True is 1 in one system and T in another, what should the value be once the data is consolidated from the two systems?
- The business must decide how to handle other necessary items - such as currency conversions

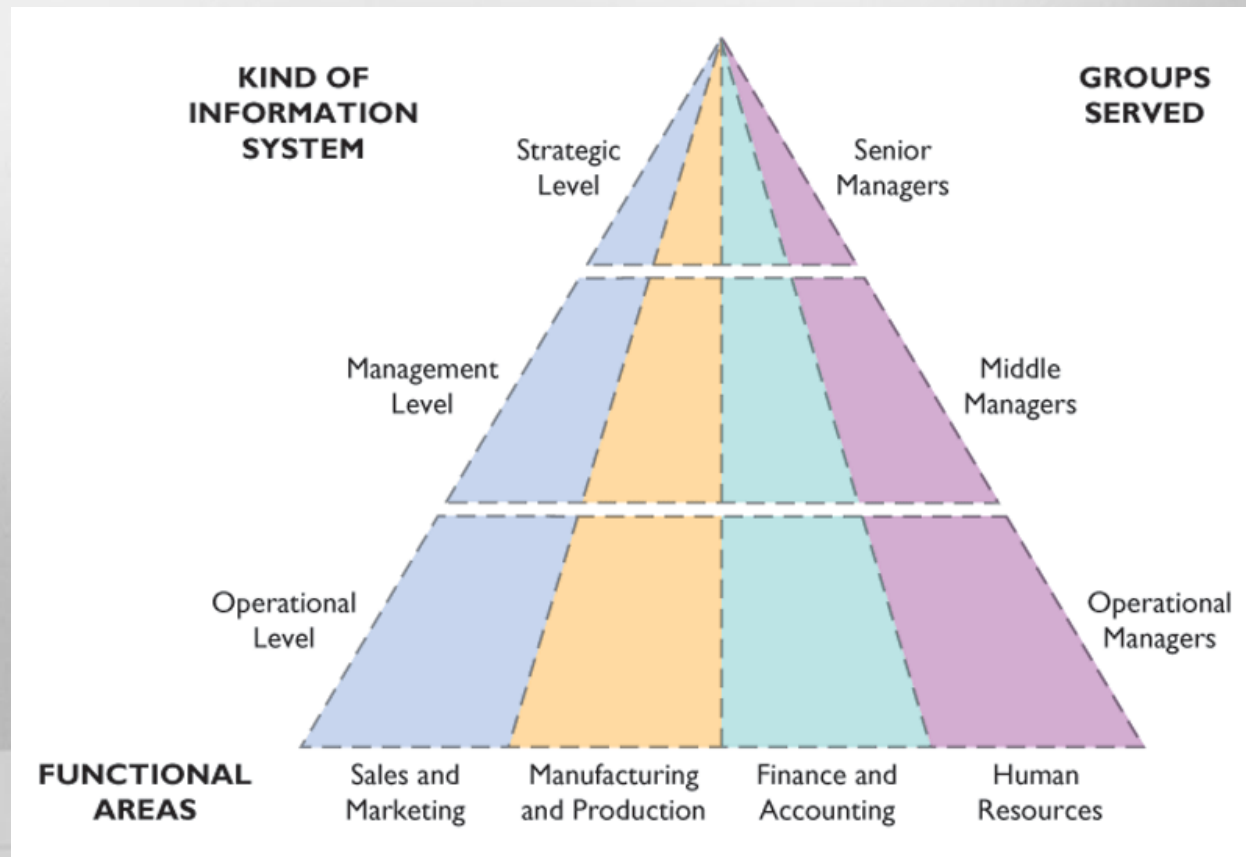
Supporting Different Types of Users

- One of the great benefits of BI is that it can support the data needs of the entire business
 - This support comes from the many different ways that users can consume BI data
- Different tools exist to support these different data needs

The Users of Business Intelligence

- Executives and business decision makers look at the business from a high level, performing limited analysis
- Analysts perform complex, detailed data analysis
- Information workers need static reports or limited analytic power
- Line workers need no analytic capabilities as BI is presented to them as part of their job

The Users of Business Intelligence



The Approaches to Consuming Business Intelligence

■ Scorecards

- Customized high-level views with limited analytic capabilities

■ Reports

- Standardized reports aimed at a large audience, with no or limited analytic capabilities

■ Analytics Applications

- Applications designed to allow complex data analysis

■ Custom Applications

- Embed BI data within an application

The Components of a Data Warehouse

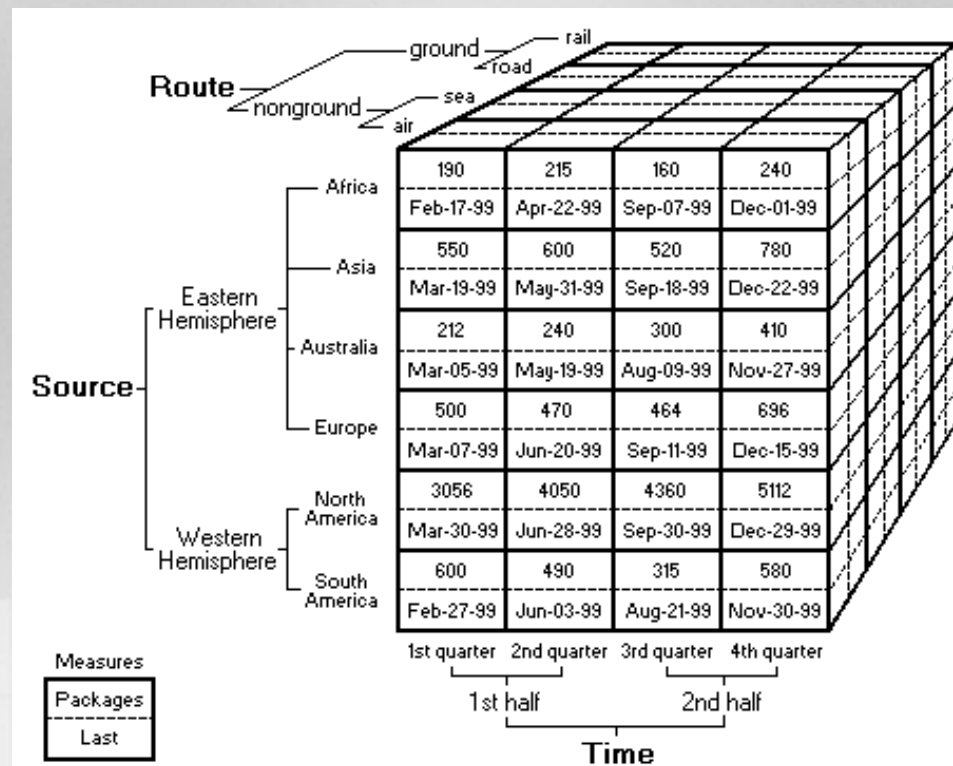
- There are several items that make up a data warehouse
 - Cubes
 - Measures
 - Key Performance Indicators
 - Dimensions
 - Attributes
 - Hierarchies

Asking a BI Question

- Humans tend to think in a multidimensional way, even if they don't realize it
- We often want to see a particular value in a certain context
 - Show me *sales by month by product* for *North America*
- “What” you want to see (sales in this case) is called a *measure*
- How you want to see it (month, product, and North America) is called a *dimension*

Cubes

- Cubes are the structures in which data is stored
- Users access data in the cubes by navigating through various dimensions



Measures

- Measures are *what* you want to see
- They are almost always numeric
- They are often additive
 - Dollar sales, unit sales, profit, expenses, and more
- Some measures are not additive
 - Date of last shipment
 - Inventory counts and number of unique customers

Dimensions

- Dimensions are *how* you want to see the data
- You usually want to see data by time, geography, product, account, employee, ...
- Dimensions are made up of *attributes* and may or may not include *hierarchies*
 - Year – Semester – Quarter – Month – Day
 - Product Category – Product Subcategory - Product

Attributes

- Attributes are individual values that make up dimensions
 - A Time dimension may have a Month attribute, a Year attribute, and so forth
 - A Geography dimension may have a Country attribute, a Region attribute, a City attribute, and so on
 - A Product dimension may have a Part Number attribute, a size attribute, a color attribute, a manufacturer attribute, and more

Hierarchies

- You can put attributes into a hierarchical structure to assist user analysis
- One of the most common functions in BI is to “drill down” to a more detailed level
- For example, Time hierarchy might be to go from Year to Quarter to Month to Day
- Another Time hierarchy might go from Year to Month to Week to Day to Hour

Summary

- The **ETL** process extracts data from source systems, transforms it and then loads it to a **data warehouse** or a data mart.
- Using **reports** and **dashboards**, **BI** looks at data as a collection of **measures** and **KPIs** viewed by dimensions.

Oracle DW/BI Products

- OBIEE – mainly based on Siebel technology.
- Oracle Hyperion Essbase

Popular BI Tools

- **Tableau**: Known for its powerful data visualization capabilities and ease of use.
- **Microsoft Power BI**: A popular choice, particularly for users within the Microsoft ecosystem, offering strong integration with other Microsoft products.
- **Qlik**: Offers a comprehensive platform for self-service data visualization, guided analytics, and reporting.
- **Looker (now Looker Studio)**: Provides a semantic layer for data modeling and real-time data exploration.
- **Domo**: Known for its cloud-based platform and ease of data management.
- **Sisense**: Offers advanced analytics, data visualization, and machine learning capabilities.
- **ThoughtSpot**: Focuses on AI-powered business intelligence, enabling users to search for insights using natural language.